Thai Travel Assistant

Capstone: Building LLM-Powered Applications (KMITL) · model `gemma-4-E4B-it` + `bge-m3` via the class proxy

1. Problem and users

Backpackers and students planning trips in Thailand ask concrete, checkable questions: what a ferry costs, how long a bus takes, how far an attraction is from town, and what a group will pay in total. General chat models answer these confidently from memory, with numbers that are often outdated or invented. Travel guides contain the answers but are long. For example, the Chiang Mai guide alone is about 157k characters.

This app answers such questions from **Wikivoyage guides for 15 destinations**, with a citation to the section each fact came from. It does arithmetic with a calculator rather than in the model's head, and it **says so when the guides don't cover a question** instead of guessing.

The corpus (15 guides, 709,647 characters, 1,321 chunks at the default size) has nothing in common with the course reference app's corpus, which documents the course's own software stack.

2. Design

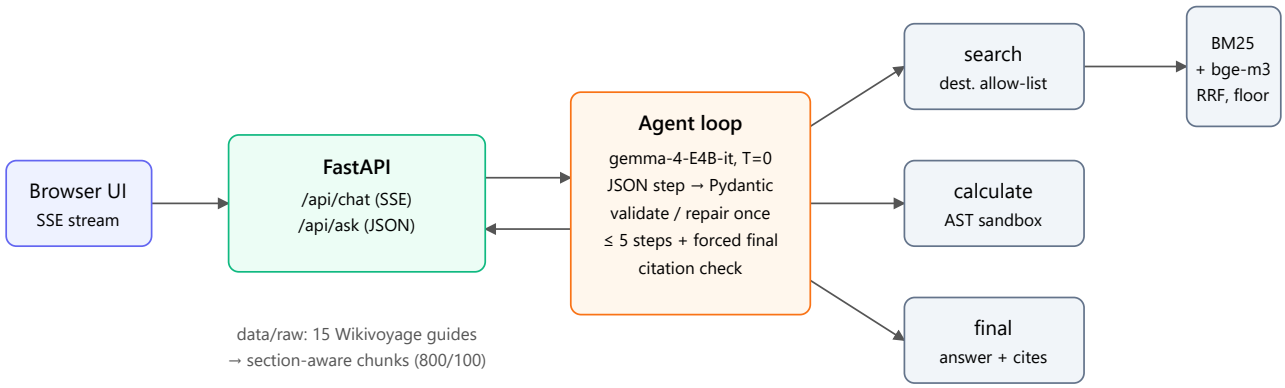


Figure 1: Architecture. Every model output is validated before anything runs.

Technique (course family)	How it is used	Why
Structured output + validation (1)	Each agent step is JSON parsed into a Pydantic discriminated union (search | calculate | final). An invalid step is sent back with the error once, then the agent gives up safely.	The proxy ignores `response_format`, and a 4B model often wraps JSON in prose. Nothing executes unless it validates.
RAG with citations (2)	Section-aware chunks with a "Destination — Section:" prefix, retrieved with hybrid BM25 + bge-m3 fused by RRF, and a cosine floor. Citations to chunks that were never retrieved are dropped.	Answers must be checkable. Hybrid retrieval catches both exact names and numbers (lexical) and paraphrases (dense).
Retrieval evaluation (3)	Hit@1, Recall@5 and MRR on 27 labelled questions, across 3 retrievers and 3 chunk sizes, plus a min_score sweep.	Every retrieval setting in the app is chosen from these numbers.
Agent loop with tools (4)	Tools: search (with an allow-listed destination filter) and calculate. The loop stops on `final` or after 5 steps,	Questions combining a fact with arithmetic need several steps. The destination filter is a

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	followed by one forced final turn. Repeated identical actions are refused.	form of query construction.
Sandboxed code execution (5)	The calculator parses expressions into an AST and evaluates only numeric literals and arithmetic operators. It never calls `eval`, and exponents are capped.	Small models make arithmetic mistakes, and running model-written code directly would be unsafe.
Served interface (6)	FastAPI with `/api/chat` streaming each agent step over SSE, plus `/api/ask` returning validated JSON.	Users can watch the agent search and calculate as it works.

Deliberately left out: text-to-SQL and MCP. The data is prose, not tables, and adding them would have meant more moving parts without a question in the test set that needs them. All configuration (key, base URL, models, retrieval settings) comes from environment variables, and every evaluated call uses `temperature=0`.

3. Evaluation set and metrics

Test set: 33 hand-written questions (`eval/testset.jsonl`): 22 single-fact lookups, 5 lookup-plus-arithmetic questions, and 6 unanswerable questions about things absent from the guides (for example a "Sukhothai metro", Ko Lipe, or casinos). A script confirmed that every answerable question's **evidence phrase** appears verbatim in its gold guide.

Relevance rule: a chunk is relevant if it comes from the gold destination and contains the evidence phrase. Because the labels don't depend on chunk IDs, the same test set scores every chunk size.

Metrics:

- Retrieval: Hit@1, Recall@5 and MRR@10.
- Answers: keyword accuracy, meaning all gold keywords are present with commas normalised. For unanswerable questions, the model must abstain to count as correct.
- An LLM-as-judge score, as a second opinion.
- Cited support: the answer is correct *and* cites a chunk containing the evidence.

Baselines:

- For retrieval: BM25 only.
- For answers: *closed-book* (no retrieval) and *single-shot RAG* (one retrieval with the raw question, no tools).

4. Results

4.1 Retrieval: retriever × chunk size (27 answerable questions)

chunk_size	retriever	hit@1	recall@5	mrr@10	misses@10
400	bm25	0.56	0.85	0.67	q06 c05
800	**bm25**	**0.63**	**0.96**	**0.76**	-
1500	bm25	0.70	0.93	0.80	-

The best configuration without a filter is **bm25** at chunk size **800**, with Recall@5 = 0.96 and MRR = 0.76. The BM25 baseline at the same chunk size gets Recall@5 = 0.96 and MRR = 0.76.

Chunk size (bm25, MRR by size): 400: 0.67, 800: 0.76, 1500: 0.80. Small chunks split a fact away from the place name it belongs to. Very large chunks blur several topics into one vector, so the relevant passage competes with unrelated text in the same chunk.

4.2 Measured improvement: the min_score floor

Not run yet: run `python -m eval.eval_retrieval` and rebuild this report.

4.3 End-to-end answers (33 questions)

Not run yet: run `python -m eval.eval_answers` and rebuild this report.

5. Failure analysis

Not run yet: run `python -m eval.eval_answers --systems agent` and rebuild this report.

6. Limitations and next steps

- **Small evaluation set.** With 33 questions (only 6 unanswerable), one question moves accuracy by about 3 points, so differences of one or two questions are within noise. Next step: grow the set to 100+ questions, including questions written by other students.
- **Stale facts.** Prices in the guides date from 2016–2025. The app reports what the guide says, not current prices. Next step: show each chunk's "updated" date next to its citation.
- **Single-hop bias.** Most questions need one passage. Questions comparing destinations, such as "cheaper to reach: Ko Tao or Ko Samet?", are untested.
- **Judge reliability.** The LLM judge is the same 4B model. A stronger judge, or a second human grader, would make that column more trustworthy.
- **English only.** The bge-m3 embeddings are multilingual, but Thai-language questions were not evaluated.

Reproduce: `python -m eval.eval_retrieval && python -m eval.eval_answers && python report/build_report.py`. Corpus text: Wikivoyage, CC BY-SA 4.0.